Assigment 2

a. Based on accuracy which model is the best one?

The best model is LDA

b. For each of the 6 other models, explain why you think it does not perform as well as the best one.

Linear regression is not inherently a classification algorithm

Polynomial Regression (deg=2) is potentially overfitting. It is slightly less robust than LDA

Naive Bayes assumes feature independence. This assumption limits its flexibility.

kNN is sensitive to the choice of k and distance metric. It can struggle when classes overlap.

QDA requires estimating more parameters. If data is limited, this can lead to less reliable parameter estimation and slightly worse accuracy.

output:

```
Model: Linear Regression
Confusion Matrix:
[[50  0  0]
 [ 0 48  2]
 [ 0  4 46]]
Accuracy: 0.9600

Model: Polynomial Regression (deg=2)
Confusion Matrix:
[[50  0  0]
 [ 0 49  1]
 [ 0  3 47]]
Accuracy: 0.9733

Model: Polynomial Regression (deg=3)
Confusion Matrix:
[[49  1  0]
 [ 1 46  3]
 [ 1  2 47]]
Accuracy: 0.9467

Model: Naive Bayes
Confusion Matrix:
[[50  0  0]
 [ 0 48  2]
 [ 0  3 47]]
Accuracy: 0.9667

Model: kNN
Confusion Matrix:
[[50  0  0]
 [ 0 47  3]
 [ 0  5 45]]
Accuracy: 0.9467

Model: LDA
Confusion Matrix:
[[50  0  0]
 [ 0 48  2]
 [ 0  1 49]]
Accuracy: 0.9800

Model: QDA
Confusion Matrix:
[[50  0  0]
 [ 0 48  2]
 [ 0  2 48]]
Accuracy: 0.9733
```